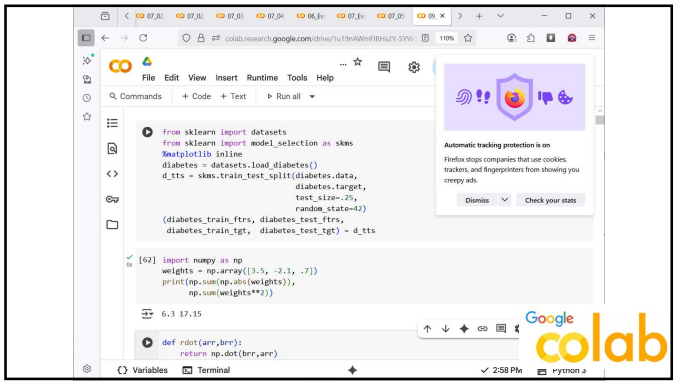
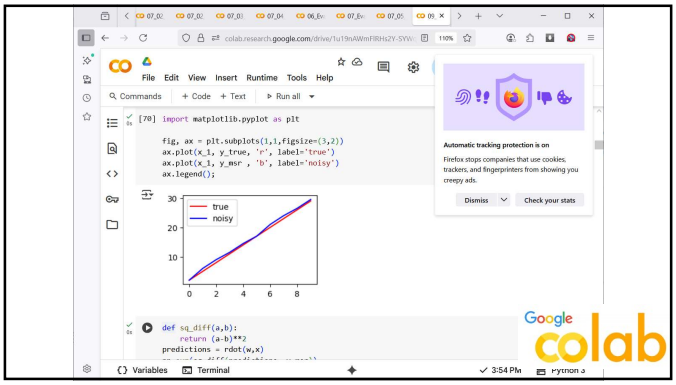
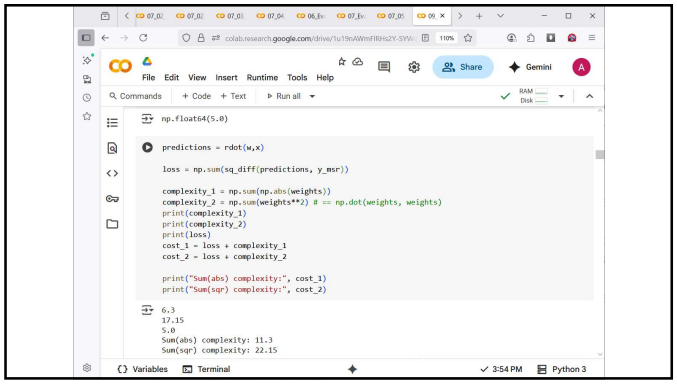
9.1. Regularizacija i regularizirana LR

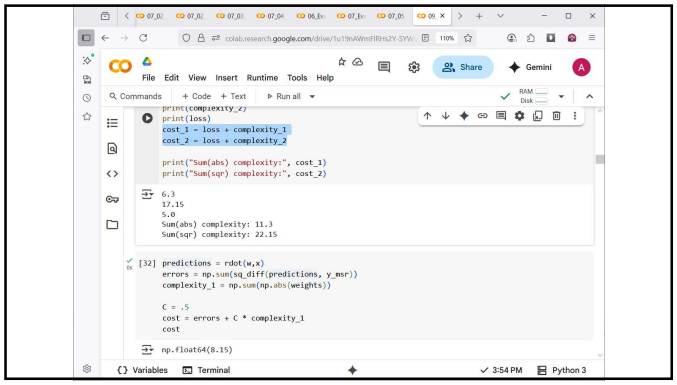
- trošak = gubitak (pogreške) + složenost
- Cost = loss + complexity
- **Regularizacija - održavanje niske složenosti i jednostavnosti modela**
- **y = w2x2 + w1x1 + w0**
- **Kako smanjiti složenost:**
  a) Postavljanjem težina na 0
  b) Postavljanjem težina na male vrijednosti ! - REGULARIZACIJA







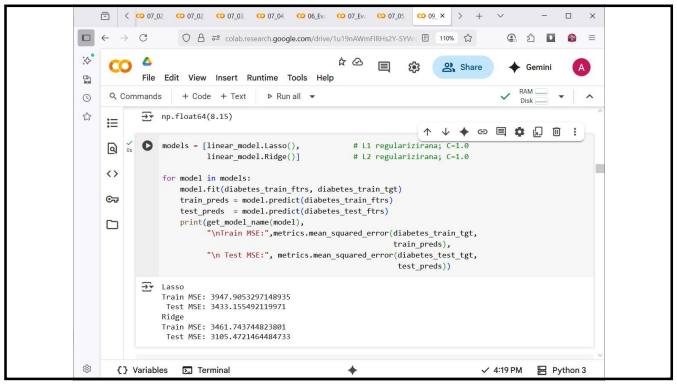
- `cost_1 = loss + complexity_1`
- `cost_2 = loss + complexity_2`



- Cost1 - L1-regularizirana regresija ili *lasso regression*
- Cost2 – L2-regularizirana regresija ili *ridge regression*

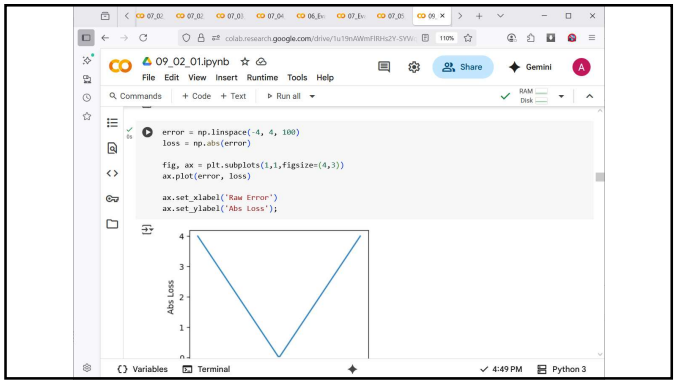
$$\text{Cost}_1 = \sum_{x,y \in D} (wx - y)^2 + C \sum_j |w_j|$$

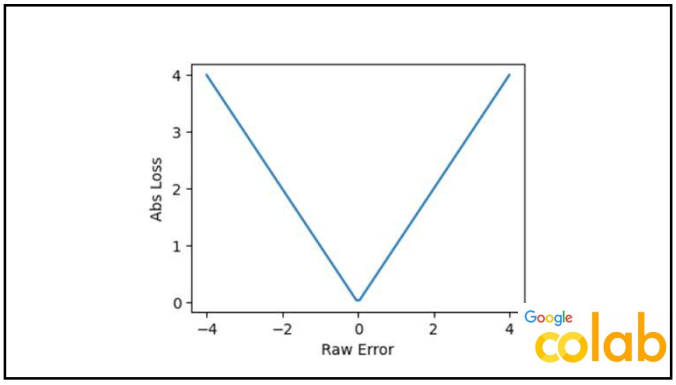
$$\text{Cost}_2 = \sum_{x,y \in D} (wx - y)^2 + C \sum_j w_j^2$$



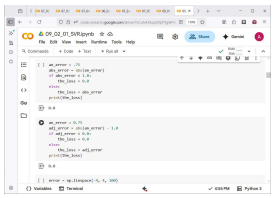
9.2. SVR – Support Vector Regression (Regresija pomoću potpornih vektora)

- trošak = gubitak + složenost
- U SVR koristimo apsolutne vrijednosti pogrešaka, ali male vrijednosti **zamenarujemo**!

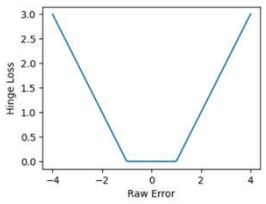




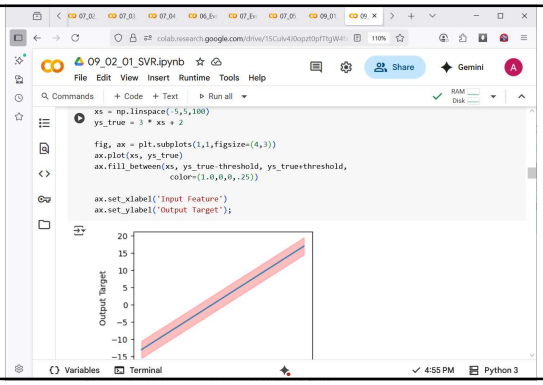
- *loss* = max(|*error*| - *threshold*; 0)

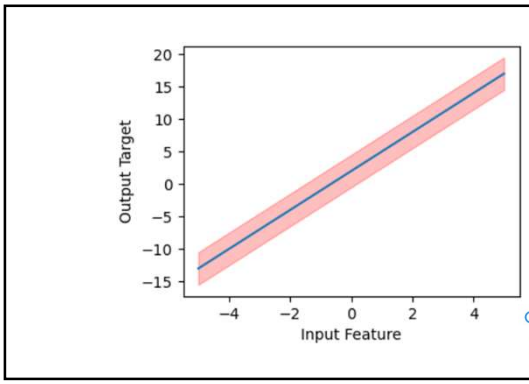


- loss = max(|error| - threshold; 0)



Google colab





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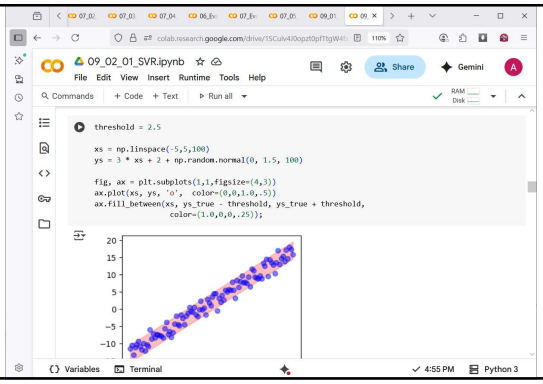
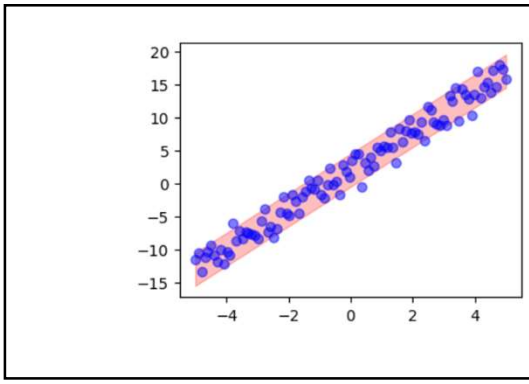
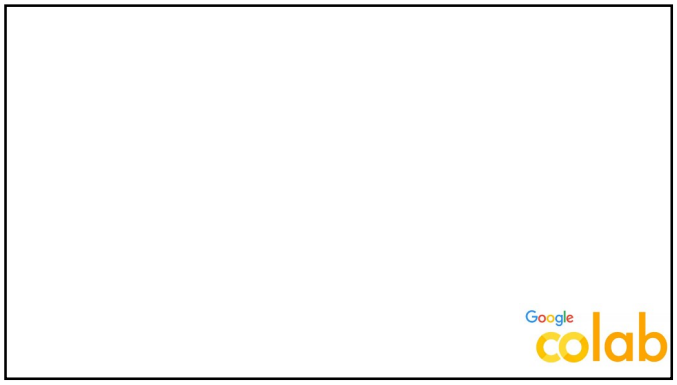
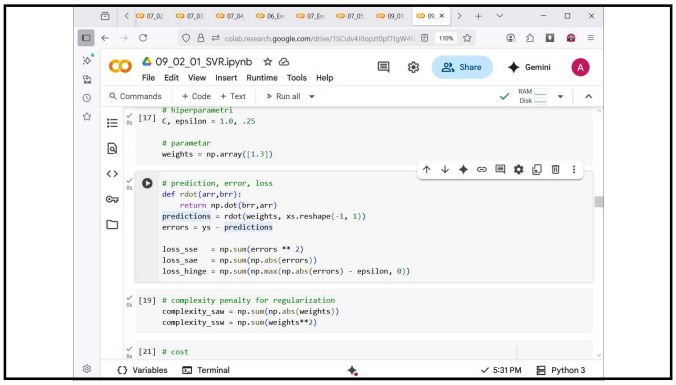
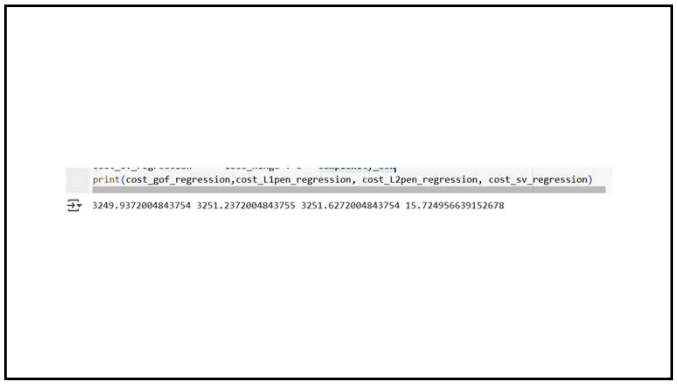






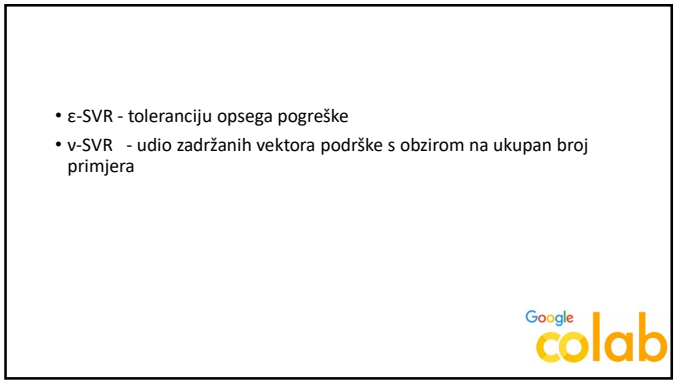


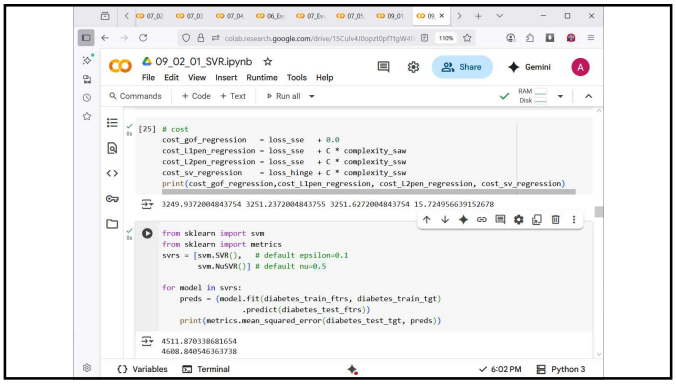


Name	Penalty	Math
GOF Linear Regression	None	$\sum_i (y_i - wx_i)^2$
Lasso	L_1	$\sum_i (y_i - wx_i)^2 + C\sum_j w_j$
Ridge	L_2	$\sum_i (y_i - wx_i)^2 + C\sum_j w_j^2$
SVR	L_1	$\sum_i \max(\lvert y_i - wx_i\rvert - \varepsilon, 0) + C\sum_j w_j$

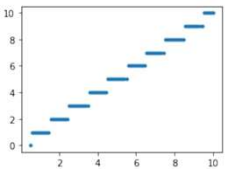
Name	Loss	Penalty
GOF LR	L_2	0
Lasso (L_1-Penalty LR)	L_2	L_1
Ridge (L_2-Penalty LR)	L_2	L_2
SVR	Hinge	L_1

- ε-SVR - toleranciju opsega pogreške
- ν-SVR - udio zadržanih vektora podrške s obzirom na ukupan broj primjera

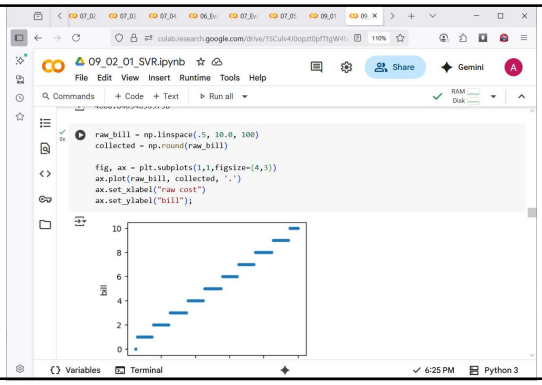


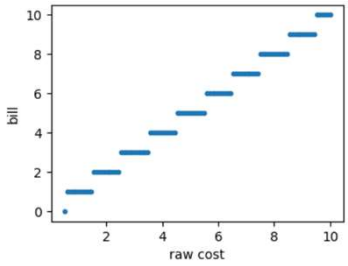


9.3 Djelomično konstantna regresija



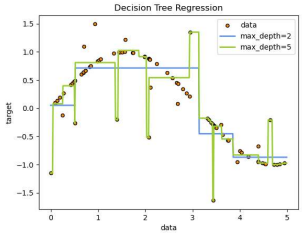
Google colab





- Sklearn dokumenti govore o tome kako implementirati prilagođene modele:
  a) Budući da implementiramo regresor, naslijedit ćemo BaseEstimator i RegressorMixin.
  b) Nećemo ništa učiniti s argumentima modela u _init__.
  c) Definirat ćemo metode fit(X,y) i predict(X)
  d) Možemo koristiti check_X_y i check_array za provjeru jesu li argumenti važeći za sklearn

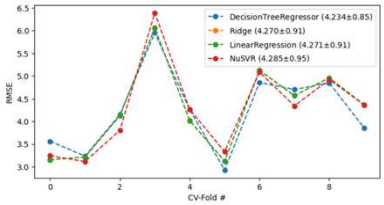
9.3. Stabla regresije



Google colab

DecisionTreeRegressor 1 4341
DecisionTreeRegressor 3 3593
DecisionTreeRegressor 5 4312
DecisionTreeRegressor 10 5190

9.4. Usporedba regresora



Google colab